

Supplementary Material

S1 – Mouse brain

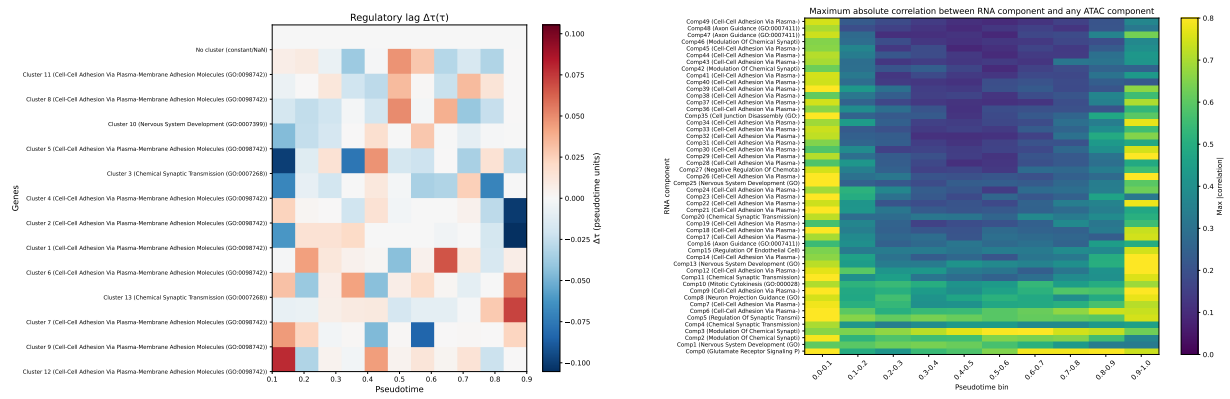


Figure 1: Mouse brain: $\Delta\tau$ heatmap (left) and ATAC-RNA correlation U-shape (right).

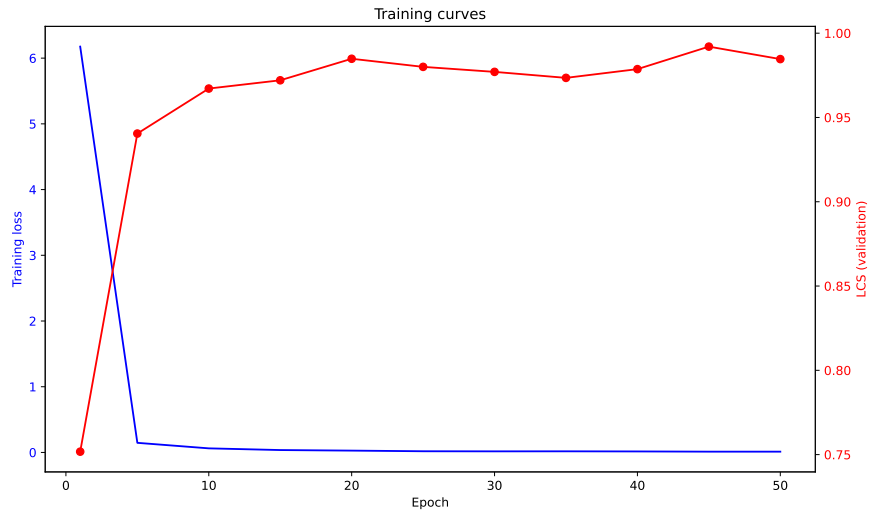


Figure 2: Mouse brain: training curves (reconstruction loss + LCS).

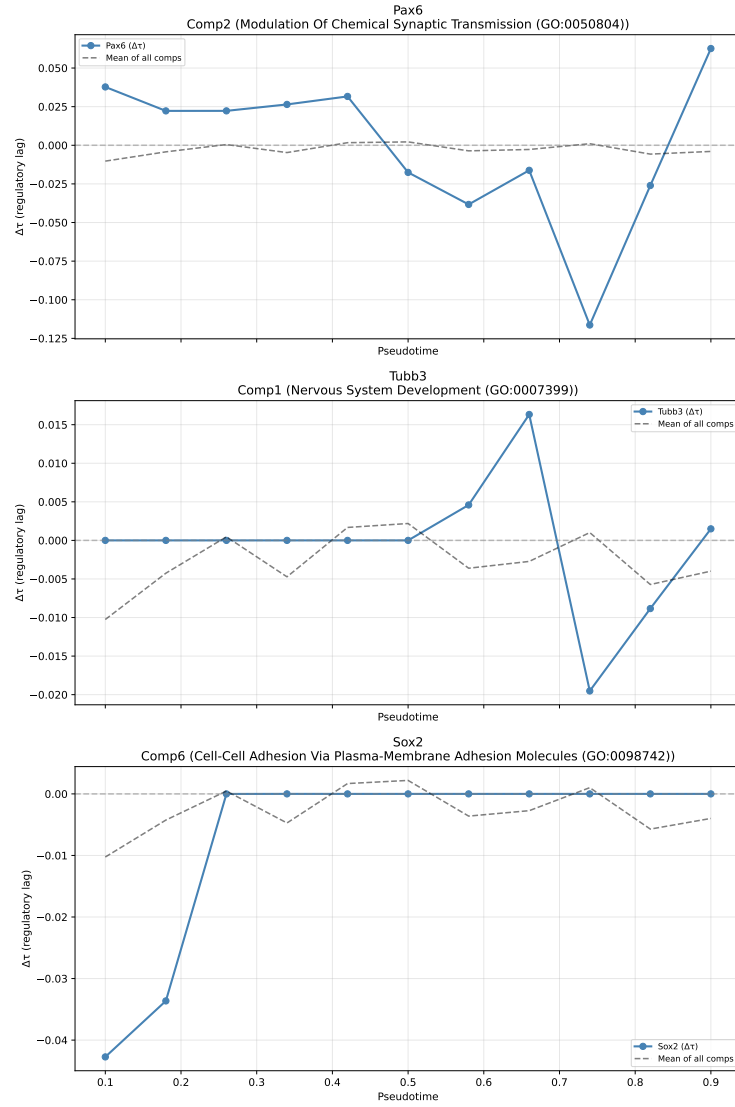


Figure 3: Mouse brain: marker $\Delta\tau$ profiles (Pax6, Tubb3, Sox2).

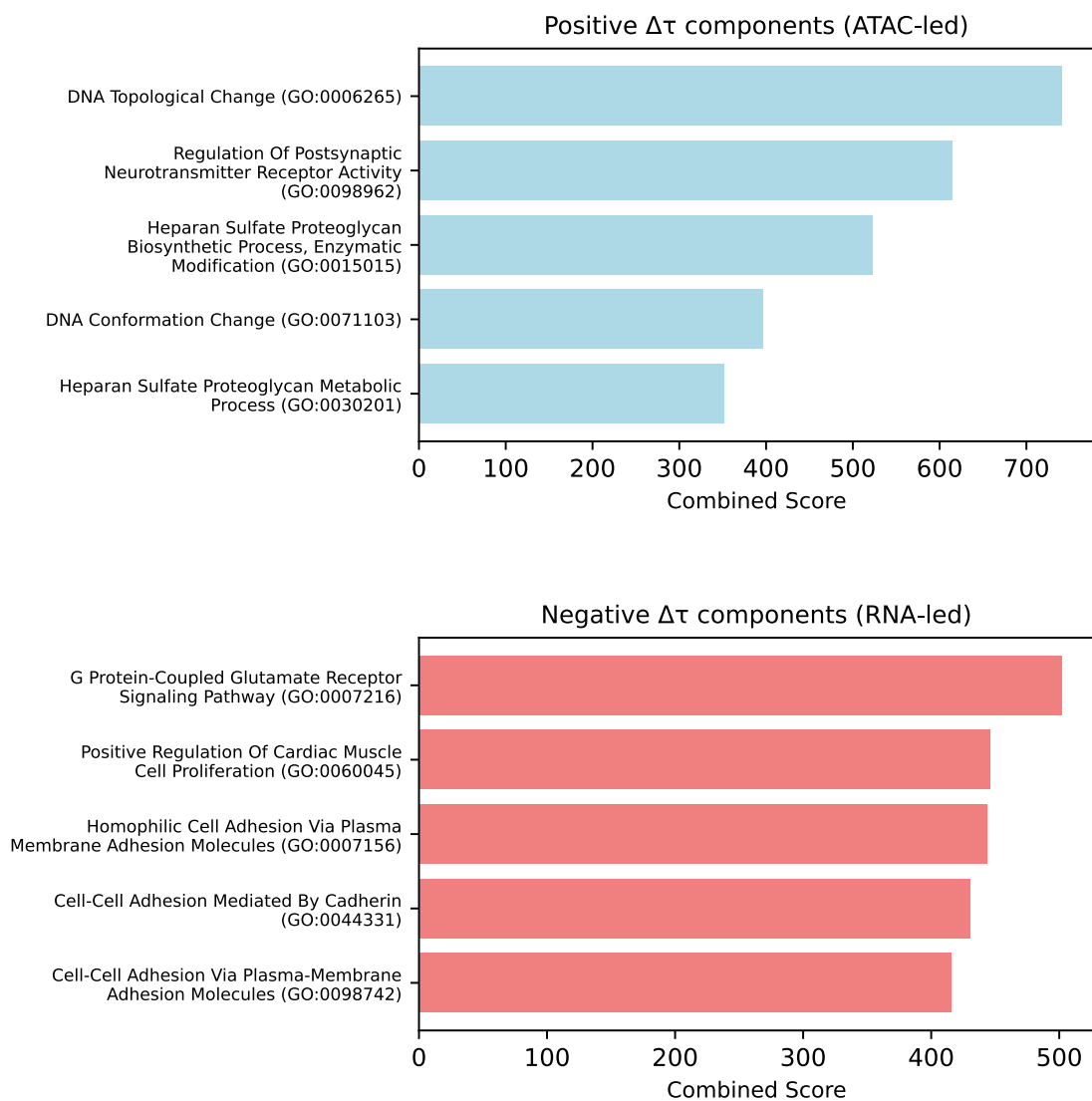


Figure 4: Mouse brain: Gene-program GO enrichment.

S2 – Human brain

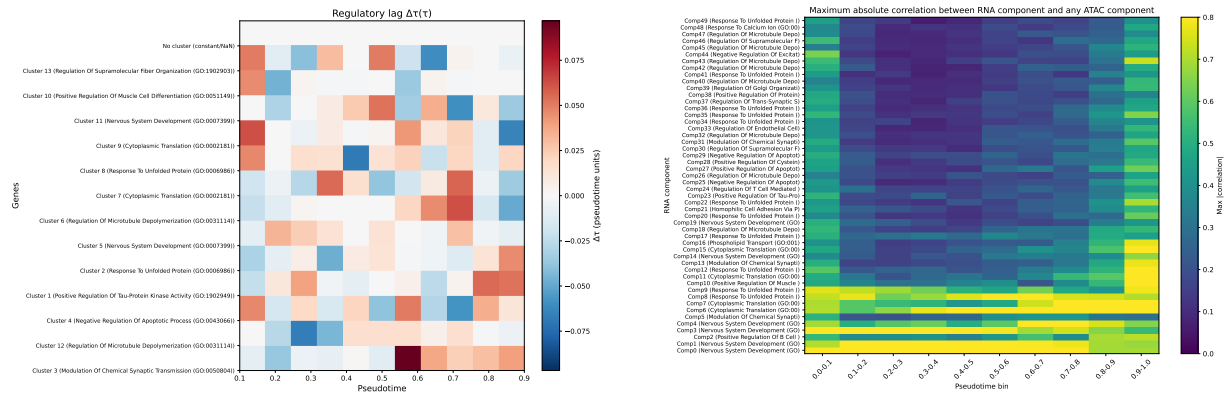


Figure 5: Human brain: $\Delta\tau$ heatmap (left) and ATAC-RNA correlation U-shape (right).

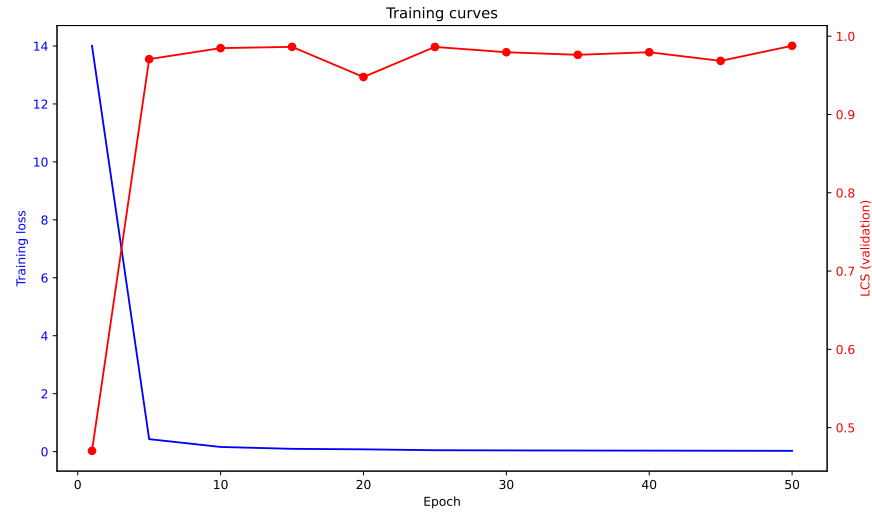


Figure 6: Human brain: training curves (reconstruction loss + LCS).

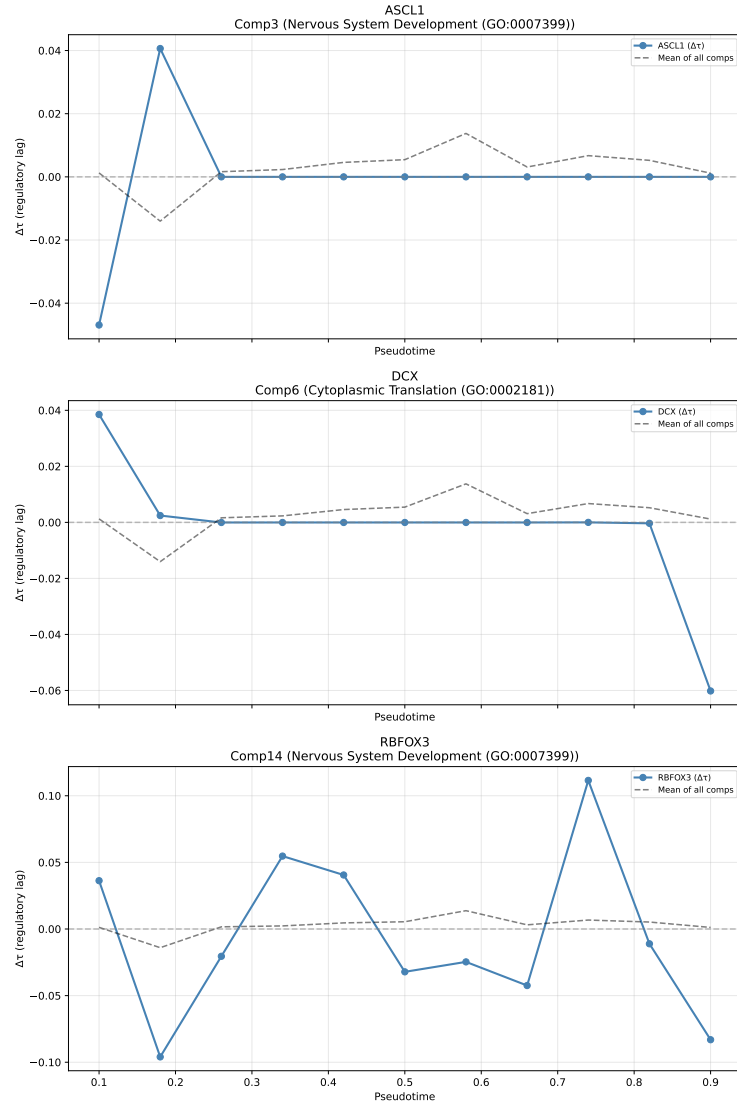


Figure 7: Human brain: marker $\Delta\tau$ profiles (ASCL1, DCX, RBFOX3).

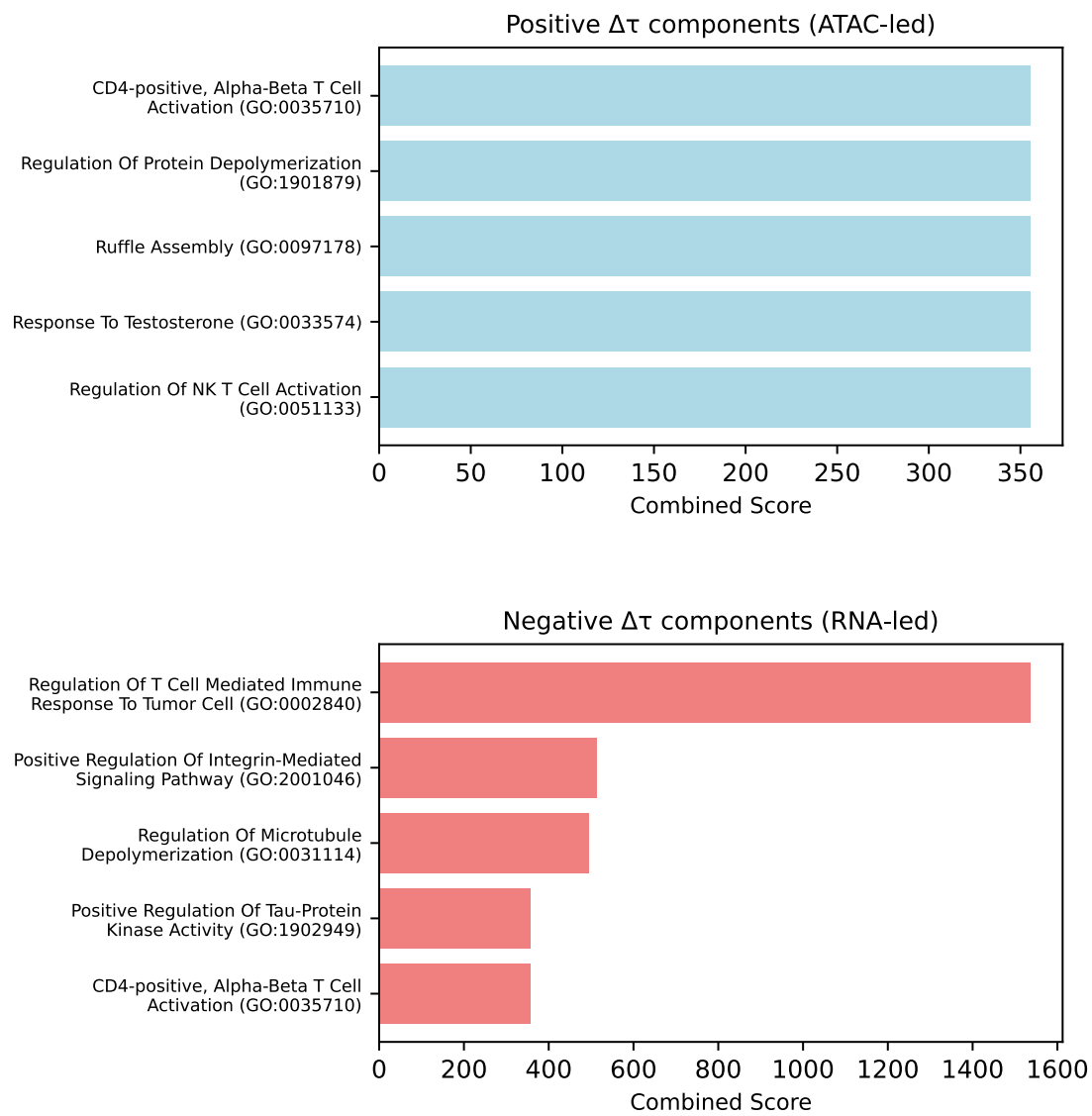


Figure 8: Human brain: Gene-program GO enrichment.

S3 – Human PBMC

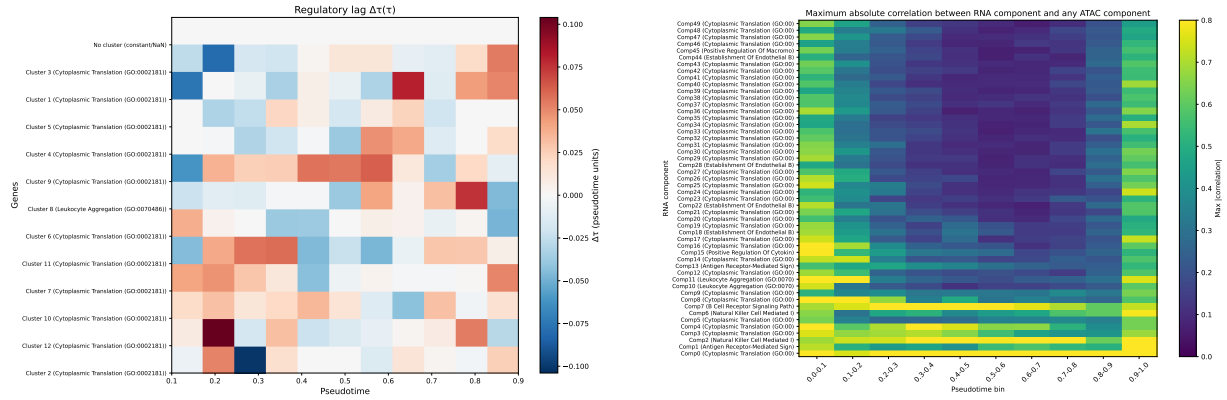


Figure 9: Human PBMC: $\Delta\tau$ heatmap (left) and ATAC-RNA correlation U-shape (right).

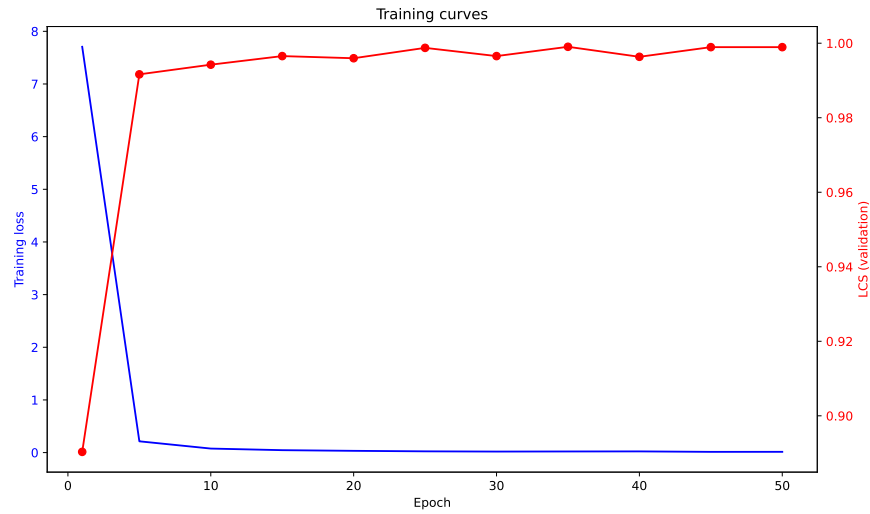


Figure 10: Human PBMC: training curves (reconstruction loss + LCS).

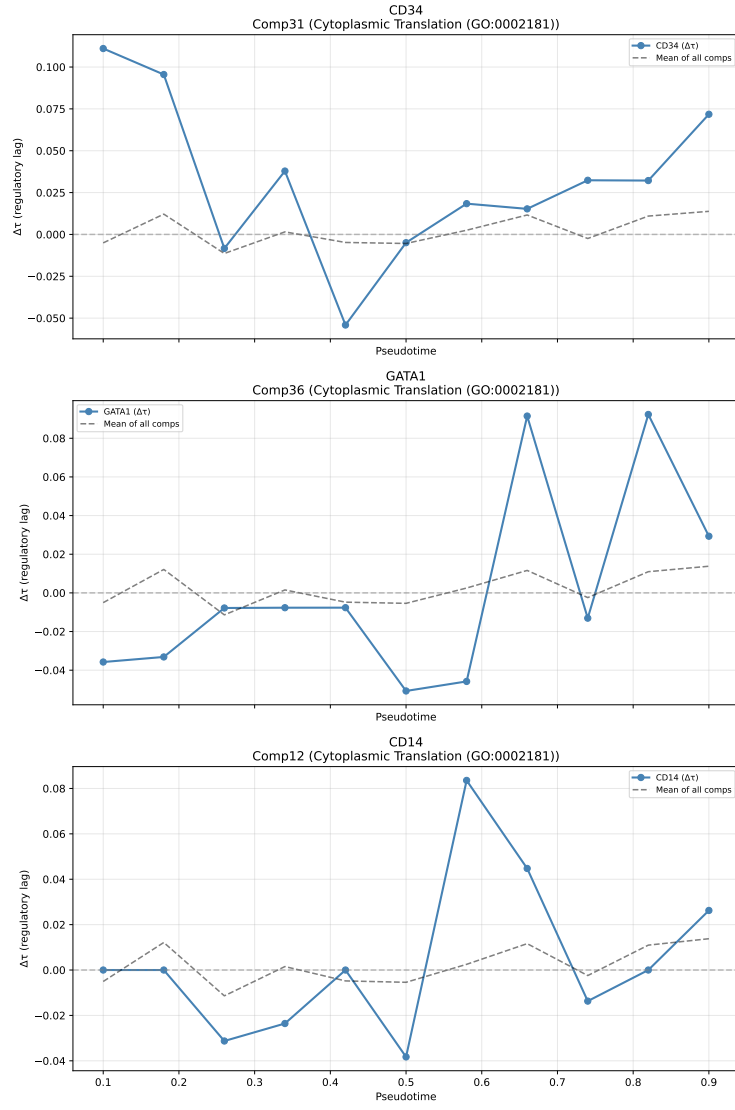


Figure 11: Human PBMC: marker $\Delta\tau$ profiles (CD34, GATA1, CD14).

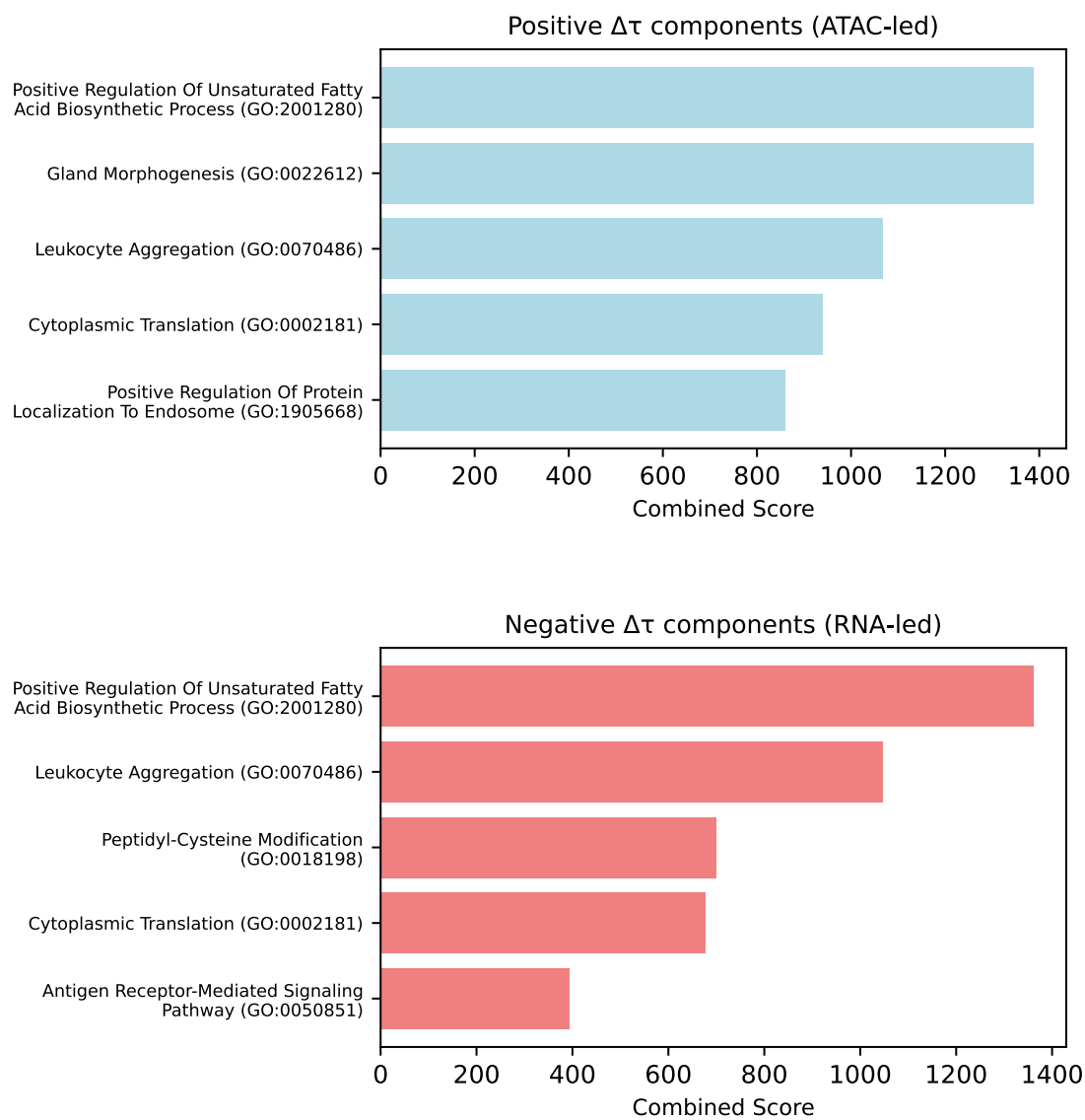


Figure 12: Human PBMC: Gene-program GO enrichment.

S4 – Mouse kidney

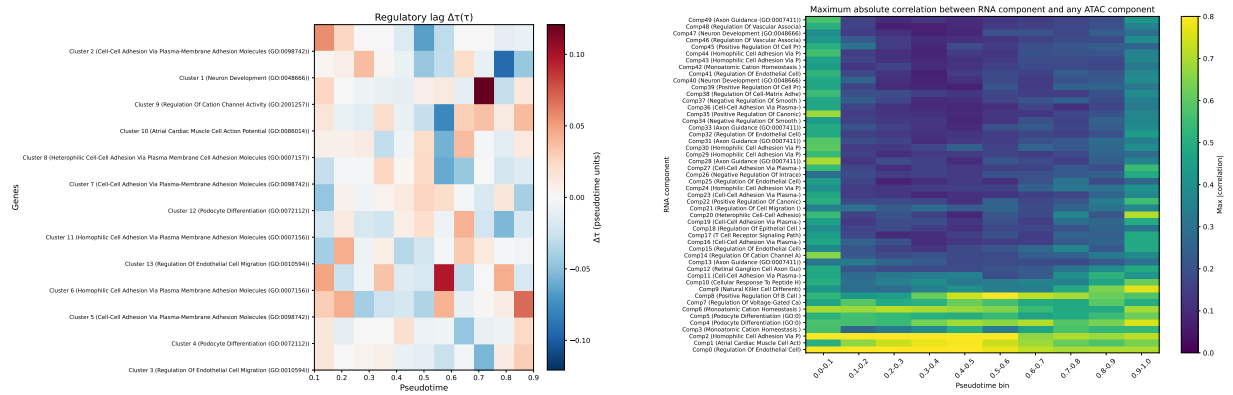


Figure 13: Mouse kidney: $\Delta\tau$ heatmap (left) and ATAC-RNA correlation U-shape (right).

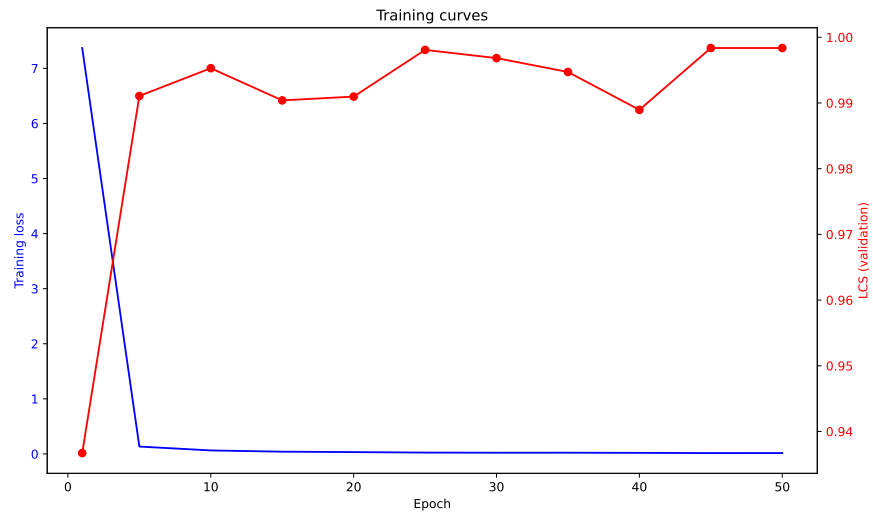


Figure 14: Mouse kidney: training curves (reconstruction loss + LCS).

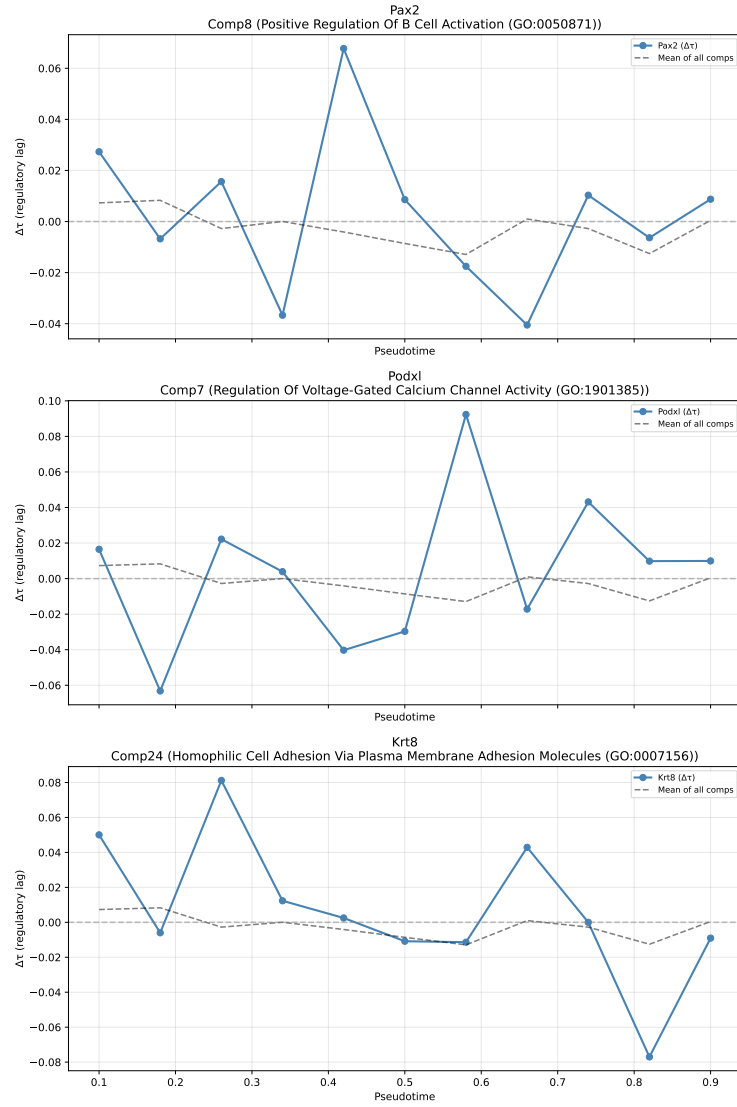


Figure 15: Mouse kidney: marker $\Delta\tau$ profiles (Pax2, Podxl, Krt8).

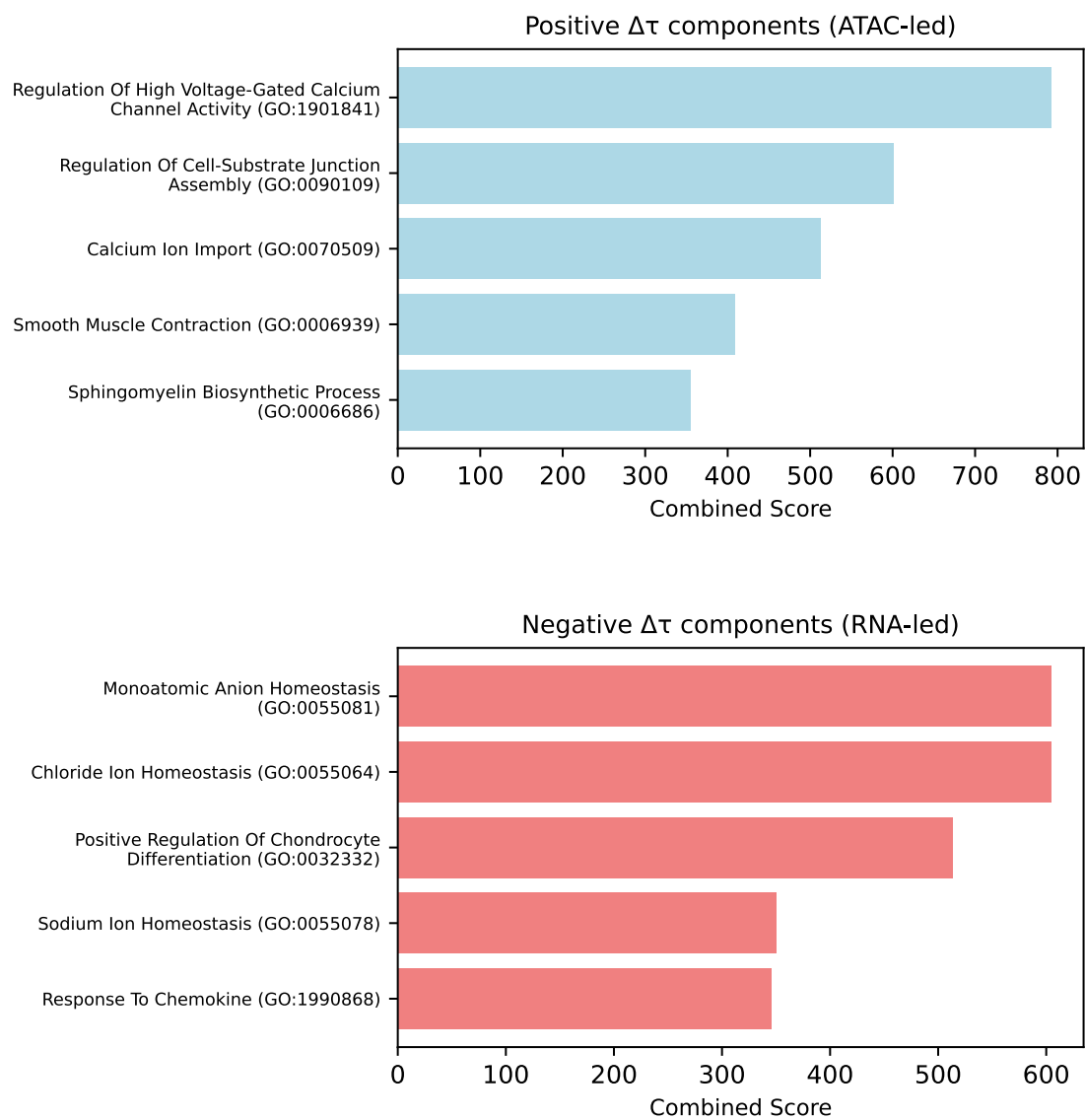


Figure 16: Mouse kidney: Gene-program GO enrichment.

S5 – LCS scatter plots (predicted vs CCF lags)

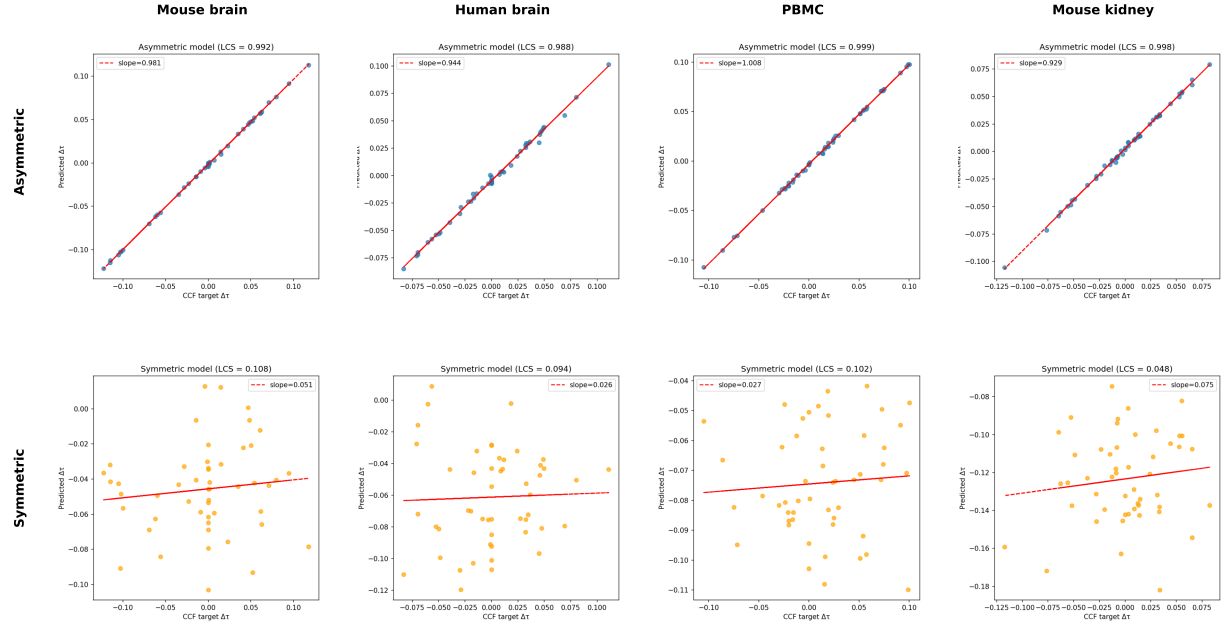


Figure 17: LCS scatter plots for all four datasets. Top row: asymmetric model; bottom row: symmetric model. Red dashed line shows the linear fit; slope and LCS are annotated.

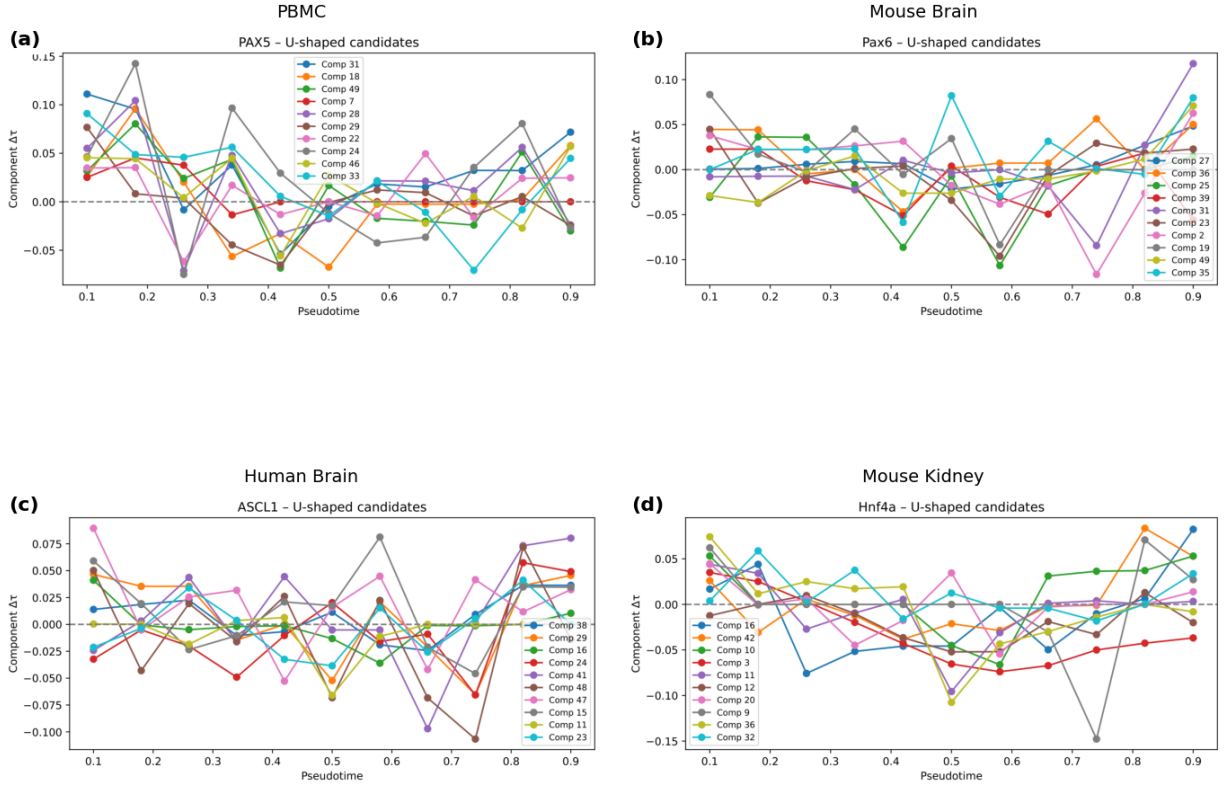


Figure 18: Biphasic regulatory lag profiles for statistically significant components in all four tissues (permutation $p < 0.05$). Each panel shows the component-level $\Delta\tau$ across pseudotime windows; a dashed horizontal line marks the ATAC-led / RNA-led boundary. The corresponding ATAC–RNA correlation U-shapes and permutation p-values are listed in Supplementary Table ST5.

Supplementary Tables

ST1 – Perturb-seq $\delta\Delta\tau$ validation summary

Table 1: Perturb-seq $\delta\Delta\tau$ results (one-sided Mann-Whitney U test).

Knockout	NTC cells	Perturbed cells	Target $\delta\Delta\tau$ mean	Background $\delta\Delta\tau$ mean	p-value
SMARCB1	1,144	147	3.0×10^{-4}	2.0×10^{-4}	0.0556
SMARCE1	25,125	3,394	~ 0	~ 0	0.0089

ST2 – Lag Generalization

Table 2: Held-out generalization results (80/20 stratified split).

Dataset	Train cells	Test cells	Held-out LCS (training target)	Test-set CCF LCS
Human PBMC	2,402	606	0.9885	-0.2092
Mouse brain	3,899	981	0.9484	-0.1556
Human brain	2,661	670	0.8483	-0.0252
Mouse kidney	4,224	1,062	0.9369	0.2214

ST3 – Full Model vs. Cell-state Ablated Model

Table 3: Cell-state ablation results across all four datasets.

Dataset	Full LCS	Ablated LCS	Mean slope full	Mean slope ablated	KS p -value
Mouse kidney	0.9984	0.7136	0.0404	0.0092	2.0×10^{-27}
Mouse brain	0.9920	0.8267	0.0142	0.0060	3.2×10^{-19}
Human brain	0.9877	0.9291	0.1114	0.0051	2.0×10^{-29}
Human PBMC	0.9990	0.9112	0.0026	0.0044	3.9×10^{-2}

Table 4: ST4 – Gene-level CCF validation results across all four datasets.

Dataset	TF	Gene-level CCF p	TMO p (component)	Variance ratio (gene/TMO)
Human PBMC	PAX5	6.3×10^{-40}	3.0×10^{-23}	3.80
Mouse brain	Pax6	2.9×10^{-38}	2.4×10^{-18}	4.67
Human brain	ASCL1	8.3×10^{-11}	1.0×10^{-3}	4.19
Mouse kidney	Hnf4a	1.1×10^{-14}	2.0×10^{-4}	3.68

Table 5: ST5 – Statistically significant biphasic regulatory lag components.

Dataset	Component	Template r	Permutation p	Correlation dip (early $\rightarrow$ mid $\rightarrow$ late)
Human PBMC	31	0.765	0.005	0.65 $\rightarrow$ 0.09 $\rightarrow$ 0.55
Human PBMC	18	0.757	0.004	0.69 $\rightarrow$ 0.14 $\rightarrow$ 0.53
Mouse brain	27	0.603	0.026	0.75 $\rightarrow$ 0.13 $\rightarrow$ 0.55
Mouse brain	36	0.584	0.037	0.66 $\rightarrow$ 0.12 $\rightarrow$ 0.53
Human brain	38	0.778	0.002	0.48 $\rightarrow$ 0.10 $\rightarrow$ 0.36
Human brain	29	0.680	0.005	0.50 $\rightarrow$ 0.16 $\rightarrow$ 0.48
Human brain	16	0.610	0.006	0.41 $\rightarrow$ 0.15 $\rightarrow$ 0.77
Mouse kidney	16	0.802	0.001	0.47 $\rightarrow$ 0.11 $\rightarrow$ 0.48
Mouse kidney	42	0.659	0.021	0.44 $\rightarrow$ 0.10 $\rightarrow$ 0.32
Mouse kidney	10	0.607	0.015	0.50 $\rightarrow$ 0.29 $\rightarrow$ 0.61
Mouse kidney	3	0.598	0.026	0.57 $\rightarrow$ 0.27 $\rightarrow$ 0.53
Mouse kidney	11	0.593	0.015	0.48 $\rightarrow$ 0.19 $\rightarrow$ 0.42
Mouse kidney	12	0.575	0.026	0.49 $\rightarrow$ 0.18 $\rightarrow$ 0.49